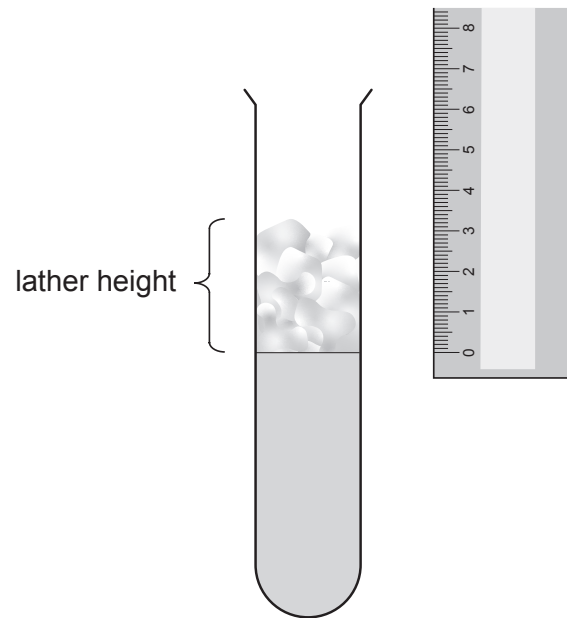


5. (a) A class measured the hardness of five different samples of water, **A**, **B**, **C**, **D** and **E**.

They added soap solution to each water sample and measured the height of the lather produced on shaking.



Their results are shown in the table.

| Water sample | Lather height (cm) |
|--------------|--------------------|
| <b>A</b>     | 1.5                |
| <b>B</b>     | 3.0                |
| <b>C</b>     | 0.5                |
| <b>D</b>     | 5.0                |
| <b>E</b>     | 2.0                |



- (i) Complete the order of hardness of the water samples, from softest to hardest. [1]

softest  $\longrightarrow$  hardest

|          |  |  |  |          |
|----------|--|--|--|----------|
| <b>D</b> |  |  |  | <b>C</b> |
|----------|--|--|--|----------|

- (ii) Calcium is one of two metal ions which cause hardness in water.

Underline the name of the other metal ion which causes hardness in water. [1]

**potassium      sodium      nickel      magnesium      tin**

- (iii) Tick (✓) **all** of the variables which must be controlled in this experiment. [2]

volume of soap solution

☐

type of water

☐

type of soap solution

☐

volume of water

☐

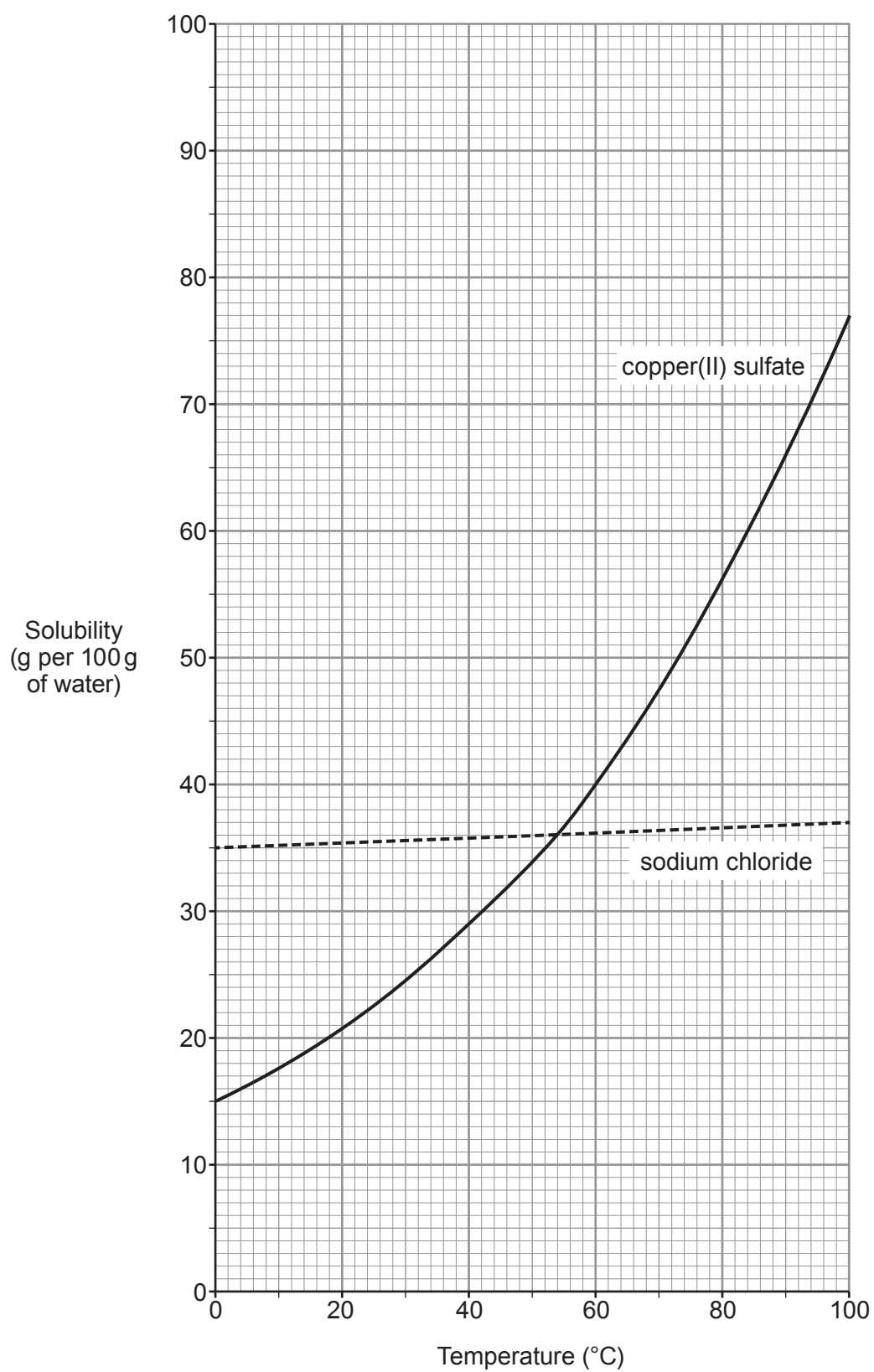
height of lather

☐

width of test tube

☐

- (b) The graph shows the solubilities of sodium chloride and copper(II) sulfate at different temperatures.



- (i) State the temperature at which the two compounds have the same solubility. [1]  
 ..... °C

- (ii) Compare the changes in solubility of the two compounds. [2]

.....

.....

.....

- (iii) Calculate the mass of copper(II) sulfate that would dissolve in **1000 g** of water at 40 °C. [2]

Mass = ..... g

- (c) Potassium sulfate,  $K_2SO_4$ , is soluble in water.

- (i) State the number of sulfur atoms in the formula  $K_2SO_4$ . [1]

.....

- (ii) Calculate the relative formula mass ( $M_r$ ) of  $K_2SO_4$ . [2]

$$A_r(K) = 39 \quad A_r(S) = 32 \quad A_r(O) = 16$$

$M_r =$  .....

